

Math 521
Week 1 Assignment

Name: Gianluca Crescenzo

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Exercise 1. Let Y have the discrete topology. Show that if $p : X \times Y \rightarrow X$ is projection onto the first coordinate, then p is a covering map.

Proof. Let $x \in X$. Then:

$$\begin{aligned} p^{-1}(X) &= X \times Y \\ &= X \times \bigsqcup_{y \in Y} \{y\} \\ &= \bigsqcup_{y \in Y} X \times \{y\}. \end{aligned}$$

Clearly the restriction $p|_{X \times \{y\}} : X \times \{y\} \rightarrow X$ is continuous. Now define $g : X \rightarrow X \times \{y\}$ by $x \mapsto (x, y)$. Clearly g is continuous, and similarly:

$$(p|_{X \times \{y\}} \circ g)(x) = p|_{X \times \{y\}}(x, y) = x.$$

Similarly

$$(g \circ p|_{X \times \{y\}})(x, y) = g(x) = (x, y).$$

Thus $p|_{X \times \{y\}}$ is a homeomorphism. Since X is an elementary neighborhood for all $x \in X$, this means p is a covering map. □

Exercise 3. Let $p : E \rightarrow B$ be a covering map; let B be connected. Show that if $p^{-1}(\{b_0\})$ has k elements for some $b_0 \in B$, then $p^{-1}(\{b\})$ has k elements for every $b \in B$.

Proof. We will first prove that for all $b, b' \in V$, where V is an elementary neighborhood, then there exists a bijection between $p^{-1}(\{b\})$ and $p^{-1}(\{b'\})$. Since V is an elementary neighborhood, there exists a family $\{W_\alpha\} \subset E$ such that $p^{-1}(V) = \bigsqcup_\alpha W_\alpha$ and $p|_{W_\alpha} : W_\alpha \rightarrow V$ is a homeomorphism for all α . For fixed α , since $p|_{W_\alpha}$ is a homeomorphism, it is bijective. Hence:

$$\text{card}(p|_{W_\alpha}^{-1}(\{b\})) = 1 = \text{card}(p|_{W_\alpha}^{-1}(\{b'\})).$$

In particular:

$$\text{card}(p^{-1}(\{b\}) \cap W_\alpha) = \text{card}(p^{-1}(\{b'\}) \cap W_\alpha).$$

Hence:

$$\text{card} \left(\bigsqcup_{\alpha} p^{-1}(\{b\} \cap W_{\alpha}) \right) = \text{card} \left(\bigsqcup_{\alpha} p^{-1}(\{b'\} \cap W_{\alpha}) \right).$$

Finally, we have:

$$\text{card}(p^{-1}(\{b\})) = \text{card}(p^{-1}(\{b'\}))$$

Since $p^{-1}(\{b\})$ and $p^{-1}(\{b'\})$ have equal cardinalities, there exists a bijection between them.

We can now begin the proof of our exercise. Define:

$$U_k := \{b \in B \mid \text{card}(p^{-1}\{b\}) = k\}.$$

We will show that the set U_k is both open and closed in B . Let $b \in U_k$. Then $b \in B$, so there exists an elementary neighborhood V of b . Let $b' \in V$. By our above lemma, there exists a bijection between $p^{-1}(\{b\})$ and $p^{-1}(\{b'\})$. Hence $\text{card}(p^{-1}(\{b\})) = \text{card}(p^{-1}(\{b'\}))$, which means $b' \in U_k$. This means $b \in V \subseteq U_k$ for some open set V , hence U_k is open. Likewise, for any $b \in B \setminus U_k$, there is an elementary neighborhood V containing b , which must then be contained in $B \setminus U_k$ since the cardinality of $p^{-1}(b')$ must be different for each $b' \in V$. Thus $B \setminus U_k$ is open. Since U_k is both open and closed in B , we conclude that $U_k = B$. Hence $p^{-1}(b)$ has k elements for every $b \in B$. $\square$

Exercise 6. Let $p : E \rightarrow B$ be a covering map. If B is compact and $p^{-1}(b)$ is finite for each $b \in B$, then E is compact.

Proof. Let $\{U_i\}_{i \in I}$ be an open cover of E . For each $b \in B$, choose an elementary neighborhood U_b such that $p^{-1}(U_b)$ is entirely contained in an element of our open cover. Then $\{U_b\}_{b \in B}$ is an open cover of B . Since B is compact, there exists points $b_1, \dots, b_n \in B$ such that $\{U_{b_1}, \dots, U_{b_n}\}$ covers B . For each i , since U_{b_i} is an elementary neighborhood, $p^{-1}(U_{b_i})$ is a disjoint union of its slices. Moreover, the number of slices is finite by hypothesis. By construction, each slice of $p^{-1}(U_{b_i})$ lies in some element of our cover $\{U_i\}_{i \in I}$. Hence there is a finite subcollection of $\{U_i\}_{i \in I}$ which covers $p^{-1}(U_{b_i})$. Since $E = \bigsqcup_{i=1}^n p^{-1}(U_{b_i})$, we can conclude E admits a finite subcover. Thus E is compact. $\square$

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Exercise 4. Consider the covering map $p : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}^2 \setminus \{0\}$ of Example 6 of §53. Find liftings of the paths:

$$\begin{aligned} f(t) &= (2 - t, 0) \\ g(t) &= ((1 + t) \cos 2\pi t, (1 + t) \sin 2\pi t) \\ h(t) &= f * g. \end{aligned}$$

Sketch these paths and their liftings.

Proof. For the function f , consider the following two equations:

$$\begin{aligned} y \cos 2\pi t &= 2 - t \\ y \sin 2\pi t &= 0. \end{aligned}$$

Solving for x and y gives:

$$\begin{aligned}x &= 0 \\y &= 2 - t.\end{aligned}$$

Then $\tilde{f}(t) := (0, 2 - t)$ is a lifting of f . Similarly for the function g , we consider the following two equations:

$$\begin{aligned}y \cos 2\pi t &= (1 + t) \cos 2\pi t \\y \sin 2\pi t &= (1 + t) \sin 2\pi t.\end{aligned}$$

Solving for x and y gives:

$$\begin{aligned}x &= t \\y &= 1 + t.\end{aligned}$$

Hence $\tilde{g}(t) := (t, 1 + t)$ is a lifting of g . For our final lifting, set $\tilde{h} := \tilde{f} * \tilde{g}$. Since $\tilde{f}(1) = (0, 1)$ and $\tilde{g}(0) = (0, 1)$, this definition is well-defined. Moreover:

$$\begin{aligned}p \circ \tilde{h} &= p \circ (\tilde{f} * \tilde{g}) \\&= p \circ \begin{cases} \tilde{f}(2t) & 0 \leq t \leq \frac{1}{2} \\ \tilde{g}(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases} \\&= \begin{cases} p \circ \tilde{f}(2t) & 0 \leq t \leq \frac{1}{2} \\ p \circ \tilde{g}(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases} \\&= \begin{cases} f(2t) & 0 \leq t \leq \frac{1}{2} \\ g(2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases} \\&= f * g \\&= h\end{aligned} \quad \square$$

Exercise 6. Consider the maps $g, h : S^1 \rightarrow S^1$ given by $g(z) = z^n$ and $h(z) = \frac{1}{z^n}$. Compute the induced homomorphisms g_*, h_* of the infinite cyclic group $\pi_1(S^1, b_0)$ into itself.

Proof. Let $b_0 = 1$. Then $g(1) = 1$ and $h(1) = 1$. Hence both maps induce homomorphisms:

$$\begin{aligned}g_* : \pi_1(S^1, 1) &\rightarrow \pi_1(S^1, 1) \\h_* : \pi_1(S^1, 1) &\rightarrow \pi_1(S^1, 1).\end{aligned}$$

Let α be the loop in S^1 given by $\alpha(t) := e^{2\pi i t}$. This is the generator of $\pi_1(S^1, 1)$, hence g_* and h_* are determined entirely by their values on $[\alpha]$. Define $p : \mathbb{R} \rightarrow S^1$ by $p(s) := e^{2\pi i s}$. Then:

$$\begin{aligned}(g \circ \alpha)(t) &= g(e^{2\pi i t}) = e^{2\pi i n t} = p(nt), \\(h \circ \alpha)(t) &= h(e^{-2\pi i t}) = e^{-2\pi i n t} = p(-nt).\end{aligned}$$

From this, $g \circ \alpha$ lifts to the path $\tilde{\alpha}_g(t) = nt$, with $\tilde{\alpha}_g(0) = 0$ and $\tilde{\alpha}_g(1) = n$. Similarly, $h \circ \alpha$ lifts to the path $\tilde{\alpha}_h(t) = -nt$, with $\tilde{\alpha}_h(0) = 0$ and $\tilde{\alpha}_h(1) = -n$. Thus:

$$\begin{aligned}g_*([\alpha]) &= [g \circ \alpha] = n[\alpha] \\h_*([\alpha]) &= [h \circ \alpha] = -n[\alpha].\end{aligned} \quad \square$$

Exercise 7. Generalize the proof of Theorem 54.5 to show that the fundamental group of the torus is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$.

Proof. Consider $p \times p : \mathbb{R}^2 \rightarrow T^2$ (where T^2 is assumed to be a subset of $\mathbb{C} \times \mathbb{C}$) given by $(p \times p)(x, y) = (e^{2\pi i x}, e^{2\pi i y})$. Since p is a covering map, $p \times p$ is also a covering map. Let $(0, 0)$ be the basepoint of $\mathbb{R}^2$ and $(p \times p)(0, 0) = (1, 1)$ the basepoint of T^2 . Then:

$$\begin{aligned} (p \times p)^{-1}(1, 1) &= \{(x, y) \in \mathbb{R}^2 \mid (p \times p)(x, y) = (1, 1)\} \\ &= \{(x, y) \in \mathbb{R}^2 \mid (e^{2\pi i x}, e^{2\pi i y}) = (1, 1)\} \\ &= \mathbb{Z} \times \mathbb{Z}. \end{aligned}$$

Since $\mathbb{R}^2$ is simply connected, the lifting correspondence $\Phi : \pi_1(T^2, (1, 1)) \rightarrow \mathbb{Z} \times \mathbb{Z}$ is a bijection. It remains to show that Φ is a group homomorphism. Let $[\alpha], [\beta] \in \pi_1(T^2, (1, 1))$ and $\tilde{\alpha}, \tilde{\beta}$ their respective lifts. Let $(n_1, n_2) := \tilde{\alpha}(1)$ and $(m_1, m_2) := \tilde{\beta}(1)$. Then by definition $\Phi([\alpha]) = (n_1, n_2)$ and $\Phi([\beta]) = (m_1, m_2)$. Now define $h : I \rightarrow \mathbb{R}^2$ by:

$$h(s) := \begin{cases} \tilde{\alpha}(2s), & 0 \leq s \leq \frac{1}{2} \\ \tilde{\beta}(2s - 1) + (n_1, n_2), & \frac{1}{2} \leq s \leq 1 \end{cases}.$$

Note that h is continuous, and:

$$((p \times p) \circ h)(s) = \begin{cases} ((p \times p) \circ \tilde{\alpha})(2s), & 0 \leq s \leq \frac{1}{2} \\ (p \times p)(\tilde{\beta}(2s - 1) + (n_1, n_2)), & \frac{1}{2} \leq s \leq 1 \end{cases}.$$

But note that $(p \times p)((x, y) + (n_1, n_2)) = (p \times p)(x, y)$ for all $(x, y) \in \mathbb{R}^2$. Hence:

$$\begin{aligned} ((p \times p) \circ h)(s) &= \begin{cases} ((p \times p) \circ \tilde{\alpha})(2s), & 0 \leq s \leq \frac{1}{2}, \\ ((p \times p) \circ \tilde{\beta})(2s - 1), & \frac{1}{2} \leq s \leq 1 \end{cases} \\ &= \begin{cases} \alpha(2s), & 0 \leq s \leq \frac{1}{2}, \\ \beta(2s - 1), & \frac{1}{2} \leq s \leq 1 \end{cases} \\ &= (\alpha * \beta)(s). \end{aligned}$$

Since $\alpha * \beta$ is a path beginning at $(1, 1)$, this means h is its unique lift beginning at $(0, 0)$. From this, we conclude that:

$$\begin{aligned} \Phi([\alpha] * [\beta]) &= \Phi([\alpha * \beta]) \\ &= h(1) \\ &= (n_1, n_2) + (m_1, m_2) \\ &= \Phi([\alpha]) + \Phi([\beta]). \end{aligned}$$

Thus $\pi_1(T^2, (1, 1)) \cong \mathbb{Z} \times \mathbb{Z}$. □

Exercise 8. Let $p : E \rightarrow B$ be a covering map, with E path connected. Show that if B is simply connected, then p is a homeomorphism.

Proof. Let $b_0 \in B$. Since B is simply connected, $\pi_1(B, b_0)$ is trivial. Since E is path connected, the lifting correspondence $\Phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0)$ is surjective. Therefore $p^{-1}(b_0)$ is a singleton. But since b_0 was arbitrary, this means every point of B has exactly one preimage. Thus p is injective. Since it is also continuous, surjective, and an open map, we conclude that p is a homeomorphism. $\square$